

Joint Confidence Regions for Rankings based on Correlated Estimates

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- Example 30 day mortality rate in hospitals, mean travel time to work by Klein
- Such tables motivate "implicit" rankings.
- Because rankings based on the observed values of $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ can vary because of sampling variability, widely understood statements of uncertainty should accompany each released ranking.
- While the margin of error gives uncertainty in the estimate $\hat{\theta}_k$ for each unit k separately, A direct assessment of the uncertainty in the estimated overall ranking would jointly involve all units and their relative standing to each other.
- If we repeat the entire process of generating all K intervals many times, at least 95 percent of the time, all K of the resulting intervals will simultaneously contain their respective true parameters $\theta_1, \dots, \theta_K$

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$$r_k = \sum_{j=1}^K I(\theta_j \leq \theta_k) = 1 + \sum_{j:j \neq k} I(\theta_j \leq \theta_k), \quad \text{for } k = 1, 2, \dots, K.$$

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- Example

k	$\hat{\theta}_k$	$\hat{r}_k$
1		
2		
3		
4		
5		

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Klein et al. (2020) assumes that $\hat{\theta}_k \sim N(\theta_k, \sigma_k^2), k = 1, 2, \dots, K$ where θ_k is unknown but σ_k^2 is known.

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Suppose that for each $k \in \{1, 2, \dots, K\}$ there exists values L_k and U_k ST

$$\theta_k \in (L_k, U_k), k = 1, 2, \dots, K, \tag{1}$$

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the range for r_k for each $k \in \{1, 2, \dots, K\}$ is as follows:

$$r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\} \tag{2}$$

where $I_k = \{1, 2, \dots, K\} \setminus \{k\}$

$$\Lambda_{Lk} = \{j \in I_k \mid U_j \leq L_k\}$$

$$\Lambda_{Ok} = \{j \in I_k \mid U_j > L_k \text{ and } U_k > L_j\}$$

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- Note that the number of elements in the range given in (2) is $|\Lambda_{Ok}| + 1$.

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1	12.3	1	(11.0, 14.3)	0		
2	17.2	2	(14.1, 21.2)			
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4	18.0	3	(17.3, 18.0)			
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1	12.3	1	(11.0, 14.3)	0	1	
2	17.2	2	(14.1, 21.2)	0	4	
3	19.1	5	(18.2, 20.1)	2	2	
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Suppose that for random quantities L_k and U_k , the intervals satisfy the following probability condition:

$$P \left[\bigcap_{k=1}^K \{ \theta_k \in (L_k, U_k) \} \right] \geq 1 - \alpha; \quad (3)$$

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$$P \left[\bigcap_{k=1}^K \{ r_k \in \{ |\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1 \} \} \right] \geq 1 - \alpha. \quad (4)$$

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- One may use the Bonferroni approach to choose t . If such an approach is used, the choice of t that would satisfy (3) is $t = z_{\alpha/2K}$.
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Joint Confidence Regions for Rankings

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Political setting — David & Legara (2015)

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Definitions and Assumptions

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Proposed methodology to compute the joint confidence region for the unordered parameters: Algorithm 1

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Joint Confidence Regions for Rankings

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We want our joint confidence region to satisfy the probability condition:

$$P\left(\hat{\theta}_k - t \cdot \sigma_k \leq \theta_k \leq \hat{\theta}_k + t \cdot \sigma_k, \forall k = 1, 2, \dots, K\right) = 1 - \alpha$$

Joint Confidence Regions for Rankings

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└ Proposed methodology to compute a joint confidence region for the ordered parameters

- We can say we are 95 percent confident that the smallest parameter up to the largest parameter falls between the calculated bounds.

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- 2: Generate $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)' \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \dots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values
- 3: Compute $\hat{\sigma}_{b(k)}^*$ using:
 - asymptotic variance definition
 - second-level bootstrap
- 4: Compute

$$t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$$

- 5: **end for**
- 6: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, \dots, t_B^*$, call this $\hat{t}$.
- 7: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ is given by
$$\mathfrak{R}_2 = [\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)}] \times [\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)}] \times \dots \times [\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)}]$$

Joint Confidence Regions for Rankings

└ Proposed methodology to compute a joint confidence region for the ordered parameters

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```

- We can say we are 95 percent confident that the smallest parameter up to the largest parameter falls between the calculated bounds.

Algorithm 2: Asymptotic Definition of Variance

- This uses results from Chen (1976) and Dudewicz (1972) to obtain an expression of the asymptotic variance of $\hat{\theta}_{(k)}$

$$\hat{\sigma}_{b(k)}^* = \sqrt{\left[\text{kth ordered value among } \left\{ \hat{\theta}_{b1}^{*2} + \sigma_1^2, \dots, \hat{\theta}_{bK}^{*2} + \sigma_K^2 \right\} \right] - \hat{\theta}_{(k)}^{*2}}$$

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Joint Confidence Regions for Rankings

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Joint Confidence Regions for Rankings

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Algorithm 3: Variance from Second-Level Bootstrap

- Second-level bootstrap algorithm

1: **for** $c = 1, 2, \dots, C$ **do**

4: **end for**

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Joint Confidence Regions for Rankings

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Joint Confidence Regions for Rankings

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Joint Confidence Regions for Rankings

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- 4: **end for**

$$\text{In } \mathfrak{R}_2, \hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_{b(k)}^* - \bar{\theta}_{\cdot(k)}^*)^2}{B - 1}}, \quad \bar{\theta}_{\cdot(k)}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{b(k)}^*$$

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5: end for
6: In  $\mathfrak{R}_2$ ,  $\hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_{b(k)}^* - \bar{\theta}_{\cdot(k)}^*)^2}{B - 1}}$ ,  $\bar{\theta}_{\cdot(k)}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{b(k)}^*$ 

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Evaluation Algorithm

For given values of Σ and $\theta_1, \theta_2, \dots, \theta_K$ (with corresponding $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ for ordered parameters)
1: **for** replications = 1, 2, ..., 5000 **do**

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Joint Confidence Regions for Rankings

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- 4: Compute the proportion of times that the condition in line 4 is satisfied and the average of T_1 , T_2 , and T_3 .
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$$T_1 = \frac{1}{K} \sum_{k=1}^K |\Lambda_{Ok}|$$

$$T_2 = \prod_{k=1}^K |\Lambda_{Ok}|$$

$$T_3 = 1 - \frac{OP}{K^2}; OP = K + \sum_{k=1}^K |\Lambda_{Ok}|$$

Measures of tightness

$$T_1 = \frac{1}{K} \sum_{k=1}^K |\Lambda_{Ok}|$$

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$$T_3 = 1 - \frac{OP}{K^2}, OP = K + \sum_{k=1}^K |\Lambda_{Ok}|$$

- Moreover, the tightness of the joint confidence region that results from Algorithm 1 is assessed using three summary measures: the arithmetic mean (T_1), geometric mean (T_2), and the metric T_3 introduced by Wright (2025)
- OP denotes the total number of occupied positions in a joint confidence region out of the total number of positions K^2 ; or the sum of the differences between the upper and lower bound of the simultaneous rank intervals added by 1, for each population k .
- Higher values of T_1 and T_2 indicate wider confidence intervals and are therefore less desirable, whereas higher values of T_3 are preferable. T_3 can range from 0, indicating no tightness, to $\frac{K-1}{K}$, implying the confidence region only contains the estimated ranking which is likely the true ranking.

Simulation settings

The following settings are considered for $K = \{10, 20, 30, 40, 50\}$. Each K has a set of corresponding variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2$. A nominal level of $1 - \alpha = 0.95$ will be used.

θ	Correlation matrix
low variability	$\mathbf{R}_{eq}, \mathbf{R}_{block}$
medium variability	$\mathbf{R}_{eq}, \mathbf{R}_{block}$
high variability	$\mathbf{R}_{eq}, \mathbf{R}_{block}$

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Correlation Structures: Equicorrelation

- This assumes that the k variables are equally correlated, i.e., that $\rho_{jk} = \rho$ where $\rho \in [-1, 1]$ for $j \neq k \in \{1, \dots, K\}$.

$$\mathbf{R}_{\text{eq}} = (1 - \rho) \mathbf{I}_K + \rho \mathbf{1}_K \mathbf{1}_K' = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}_{K \times K}$$

Joint Confidence Regions for Rankings

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• The full block correlation matrix can be expressed as

$$\mathbf{R}_{\text{block}} = \begin{bmatrix} \mathbf{R}_{eq,1} & \mathbf{C}_{12} & \cdots & \mathbf{C}_{1G} \\ \mathbf{C}_{21} & \mathbf{R}_{eq,2} & \cdots & \mathbf{C}_{2G} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{C}_{G1} & \mathbf{C}_{G2} & \cdots & \mathbf{R}_{eq,G} \end{bmatrix}_{K \times K}$$

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- Useful in the context of pre-election surveys
- In a block correlation matrix $\mathbf{R}_{\text{block}}$ with G blocks, each diagonal block represents an equicorrelation structure within group g , denoted by

$$\mathbf{R}_{eq,g} = (1 - \rho_g) \mathbf{I}_{n_g} + \rho_g \mathbf{1}_{n_g} \mathbf{1}_{n_g}'$$

where ρ_g is the within-block correlation and n_g is the number of variables in block g such that $\sum_{g=1}^G n_g = K$.

- The off-diagonal blocks capture between-block correlations, represented by $\mathbf{C}_{g'g} = \mathbf{C}_{gg'} = \rho_{gg'} \mathbf{1}_{n_g} \mathbf{1}_{n_{g'}}'$ where $g \neq g' \in \{1, \dots, G\}$

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Correlation Structures: Distance-based correlation

- Spatial dependence can be modeled using a stationary Matérn correlation function, which for two locations $\mathbf{s}_i$ and $\mathbf{s}_j$ is expressed as

$$\rho_{\text{matern}} = \frac{2^{1-\nu}}{\Gamma(\nu)} (\kappa \|\mathbf{s}_i - \mathbf{s}_j\|)^\nu K_\nu(\kappa \|\mathbf{s}_i - \mathbf{s}_j\|)$$

where $\|\cdot\|$ denotes the Euclidean distance and K_ν is the second kind of the modified Bessel function. It has a scale parameter $\kappa > 0$ and a smoothness parameter $\nu > 0$.

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Joint Confidence Regions for Rankings

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- Useful in the mean travel time from the study of Klein et al. (2020).

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- Proposed contribution: Develop a procedure to construct joint confidence intervals for the ranks and the ranked parameters when the estimates to be ranked may be correlated.

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Joint Confidence Regions for Rankings

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Joint Confidence Regions for Rankings

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